

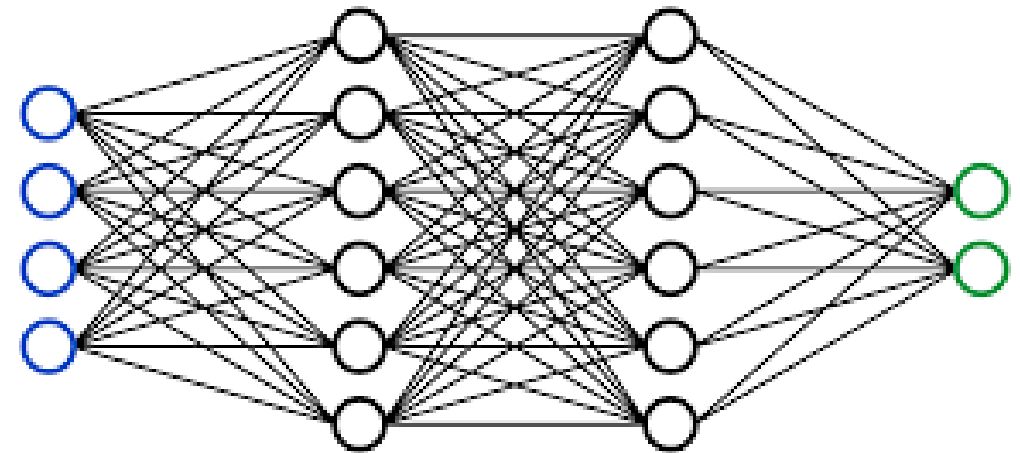
Neural Greeks for Worst-of Basket Options

Research Internship Project, Summer 2025

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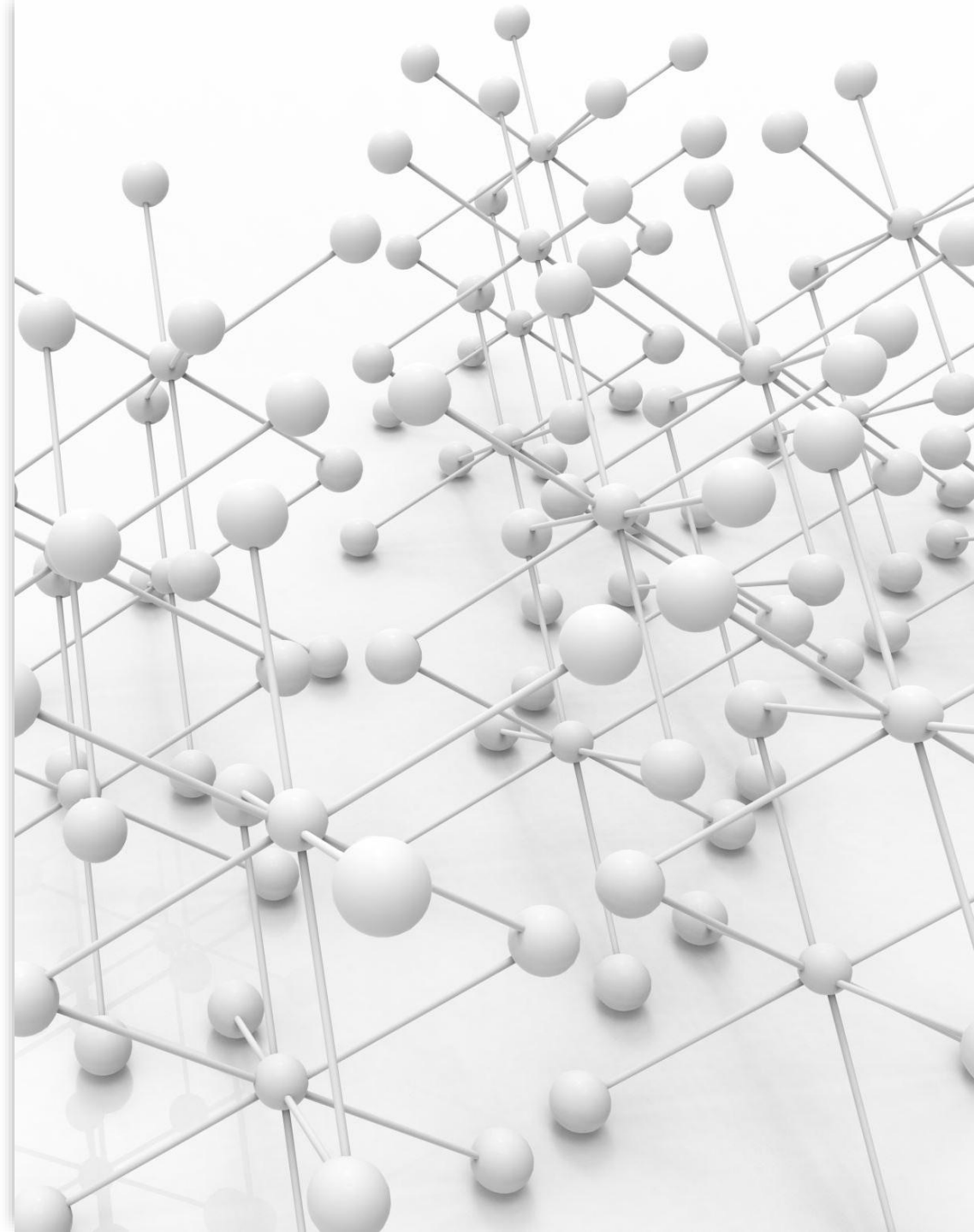
Quick Terminology

- **Worst-of Basket Option:** exotic derivative where payoff depends on the worst-performing asset in the basket.
- **Finite Differences (FD):** bump an input, re-run Monte Carlo → get Greeks (accurate but slow).
- **Adjoint Algorithmic Differentiation (AAD):** reverse-mode differentiation → all Greeks in one pass (faster, still costly).
- **Neural Networks (NNs):** machine learning models that can approximate option prices and Greeks at constant-time inference.



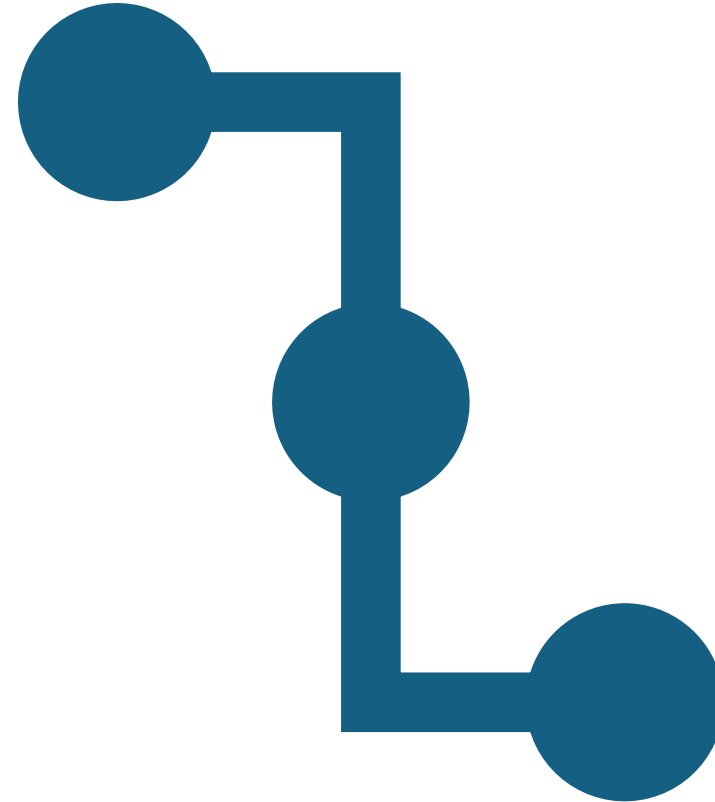
Project Overview

- **Worst-of basket options** are highly challenging to model because their payoffs exhibit *steep curvature, discontinuities, and extreme sensitivity to correlation structures*. Small changes in inputs can lead to large swings in option value and Greeks.
- In practice, **Monte Carlo simulation with Adjoint Algorithmic Differentiation (AAD)** is the *industry gold standard* for accurate pricing and Greeks. However, at portfolio scale, it is prohibitively expensive, requiring billions of paths across thousands of scenarios.
- **The research question:** *Can neural networks act as efficient surrogates — delivering the same accuracy as Monte Carlo + AAD, but at constant-time inference speeds suitable for production risk systems?*



Pipeline

1. Generate data with Monte Carlo (FD & AAD)
2. Train neural networks (ANP & DHN)
3. Compare accuracy, Greeks, and speed



Results

Model	Quantity	MAE (avg)	RMSE (avg)	R ² (avg)
ANP	Price	1.63e-03	2.38e-03	0.9999
ANP	Delta	9.54e-03	1.71e-02	0.9931
ANP	Vega	1.61e-02	2.55e-02	0.9830
DHN	Price	2.76e-03	3.52e-03	0.9999
DHN	Delta	2.07e-03	3.95e-03	0.9997
DHN	Vega	3.11e-03	5.01e-03	0.9994

- Finite Differences (FD): ~1000 seconds per 1k rows (too slow for portfolios).
- Adjoint Algorithmic Differentiation (AAD): ~322 seconds per 1k rows (better, but still costly).
- Neural Nets (ANP & DHN): ~0.002 seconds per 1k rows (constant-time, ~10⁵× faster)
- Portfolio context: Monte Carlo overall ≈ hours → Neural Nets ≈ milliseconds.

Skills Gained

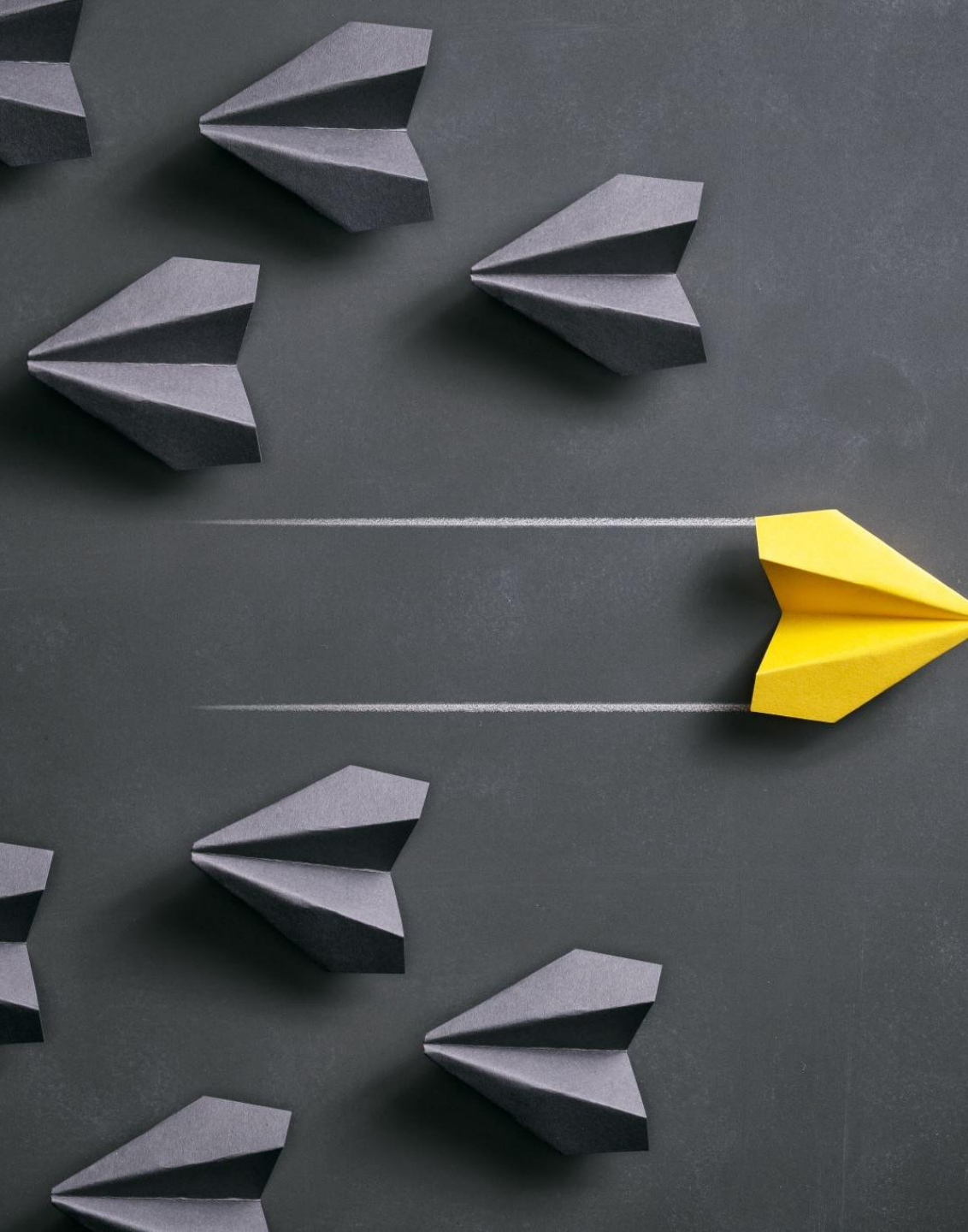
Hard Skills

- Python (GPU parallelism, NN packages)
- Differentiation & quantitative math
- Benchmarking & performance analysis

Soft Skills

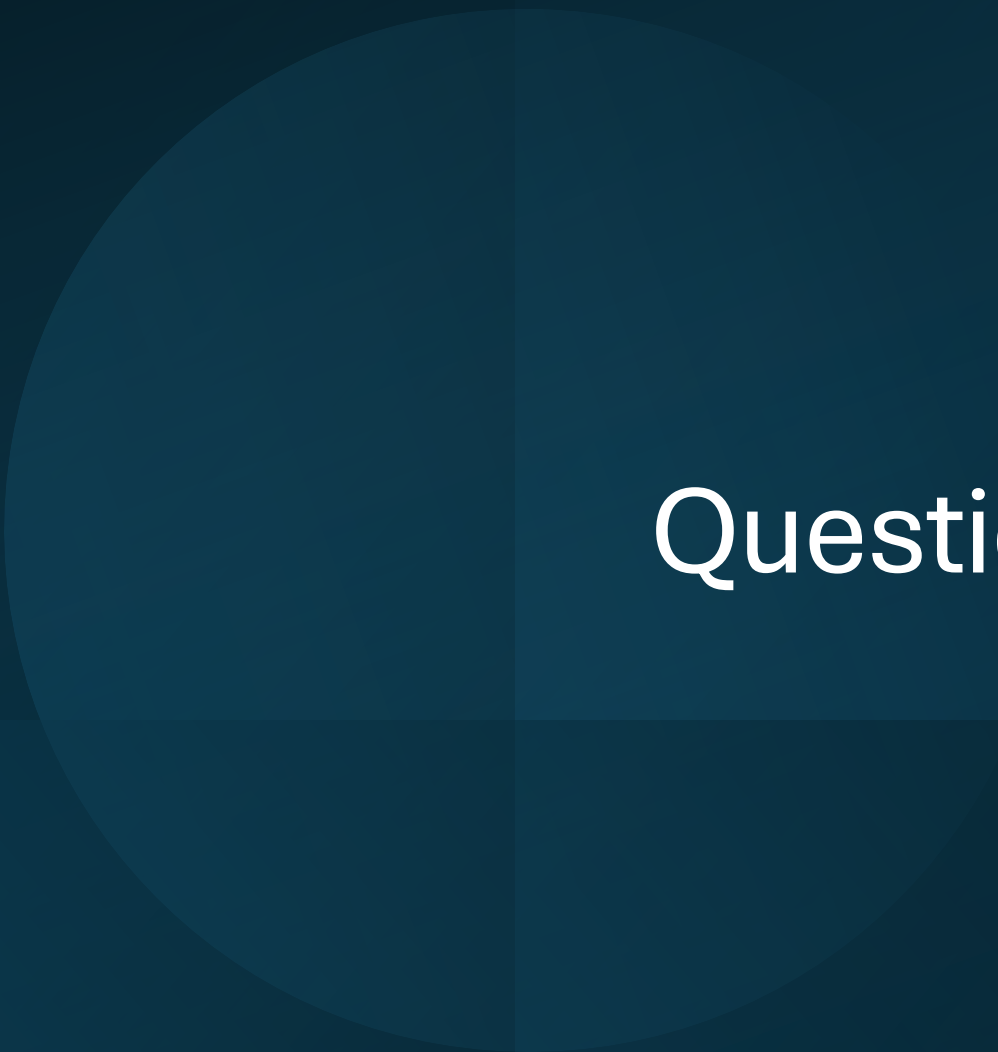
- Collaboration & meeting deadlines
- Version control (GitHub, Markdown, code documentation)





Career Goals

- Primarily interested in Risk Management
- Also open to XVA and Financial Modelling
- Most motivated when I can focus on building models



Questions / End